

Observations :-

Table No 01 :- Masses of naphthalene and biphenyl

Molar fraction of naphthalene	Mass of Naphthalene (± 0.001) /g	Mass of biphenyl (± 0.001) /g
0.16	0.137	0.863
0.20		
0.25		
0.33		
0.40		
0.50		
0.67		
0.83		

Table No 02 :- Temperature values of $\frac{1}{2}$ min intervals at different molar fractions of naphthalene.

Time (± 0.1) /s	Temperature (± 0.5) / $^{\circ}\text{C}$							
	0.16	0.20	0.25	0.33	0.40	0.50	0.67	0.83
0.0								
30.0								
60.0								
90.0								
120.0								
150.0								
180.0								
210.0								
240.0								
270.0								
300.0								
330.0								
360.0								
390.0								
420.0								
450.0								
480.0								
510.0								
540.0								
570.0								
600.0								

Table No 03 :- Temperature values for 1st trace of solid separating out

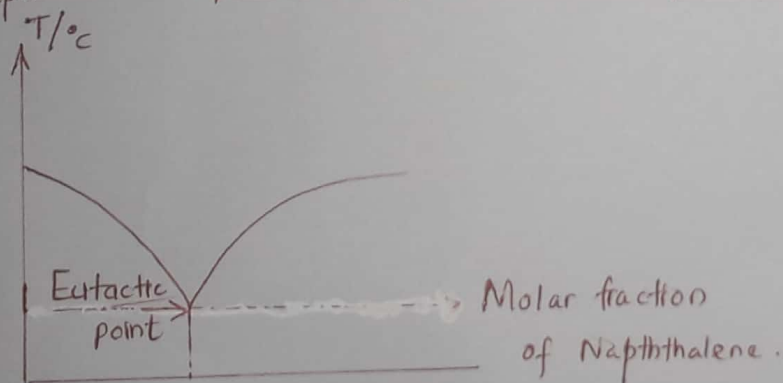
Molar fraction	Temperature of forming solid separating out (± 0.5) $^{\circ}\text{C}$
0.16	
0.20	
0.25	
0.33	
0.40	
0.50	
0.67	
0.83	

Table No 04 :- Melting point range of Biphenyl and Naphthalene

Compound	Experimental M.P (± 0.5) $^{\circ}\text{C}$	Theoretical M.P (± 0.5) $^{\circ}\text{C}$
Naphthalene		
Biphenyl		

Calculations :-

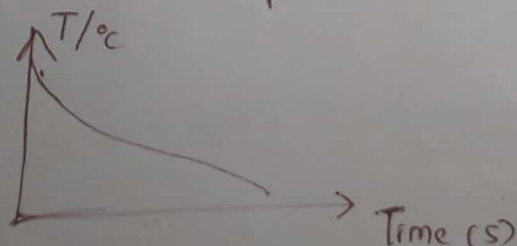
Graph No 01 :- phase diagram for Naphthalene - Biphenyl system.



From the graph;

The eutactic point of naphthalene - biphenyl system is _____ $^{\circ}\text{C}$

Graph No 02 :- Temperature vs time graph for $X_N = 0.16$



(Likewise you should draw for $X_N = 0.20, 0.25, 0.33, 0.40, 0.50, 0.67, 0.83$) $\Rightarrow$ 8 graphs

Conclusion :-

The eutactic point of naphthalene - biphenyl system is $\text{---}^{\circ}\text{C}$

Melting point of naphthalene is

Melting point of biphenyl is

Discussion .

- (1) Structure of Naphthalene and biphenyl
- (2) Naphthalene and biphenyl system. (phase diagram)
- (3) What is the eutactic point on a phase diagram?
What happen when the system reach eutactic point?
- (4) purpose of the cooling curves .
- (5) Applications of phase diagram
- (6) Errors of the experiment .

Reference .

(1)

(2)

(3)